A Reproducible CR3BP Benchmark Resource for Low-Thrust Cislunar Missed-Thrust Initialization

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Abstract

This paper presents a reproducible normalized Earth–Moon CR3BP benchmark resource for low-thrust cislunar missed-thrust initialization. The scope is preliminary algorithm benchmarking and provenance, not flight design, high-fidelity validation, full ephemeris propagation, production robust optimization, or operational targeting; all terminal-error thresholds are normalized screening tolerances. The resource archives source states, outage masks, configurations, result CSV/JSON files, generated tables and figures, cached force-model contrasts, and an artifact manifest. Under equal true trajectory-evaluation budgets, the initializer comparison does not support quantum or QAOA superiority: the configured statevector-QAOA initializer is weaker than the strongest classical initializers in the 30-seed main-method package, and optimized simulated QAOA is not statistically superior to surrogate-QUBO simulated annealing in the focused ablation. The positive evidence is narrower. A duty-cycle prior helps several samplers find refinable high-duty schedules, and the continuous-backend/provenance ladder separates selected-branch evidence, all-mask diagnostics, all-configured-mask rows, deterministic replay, and force-model stress. A new independent-midpoint Hermite–Simpson all-configured CR3BP headroom package optimizes all eight one-segment outage masks at phase time 0.4 and $a_{\max} = 0.2$: the original row reaches nominal error 0.0111 and all-configured worst error 0.0774 while retaining a max-function-evaluation caveat, while a polished warm start converges with nominal error 0.0113 and worst error 0.0779. Persisted endpoint and midpoint branch controls for both p=0.4 rows replay exactly under normalized CR3BP, and the converged polish row passes a deterministic simple bicircular solar-tidal phase sweep with maximum nominal error 0.02214 and maximum branch-phase error 0.08557. A separate cached-Horizons-derived solar-tidal replay at a representative 2026-Jan-01 epoch gives polish-row nominal error 0.01836, branch worst error 0.07422, branch pass count 8/8, and CR3BP replay delta 0.0. Under a stronger cached-Horizons Earth/Moon/Sun point-mass model at the same representative epoch, persisted controls fail direct replay with nominal error 0.38126, but independent retuning restores nominal error 0.02144 and branch worst error 0.02473 for all eight branches. A SPICE-derived replay of the same 36 already-retuned controls under compact NAIF vector caches gives nominal/branch worst errors 0.02144/0.02473, branch pass count 32/32, and maximum absolute replay delta 3.28×10^{-10} from the Horizons-retuned replay; it remains an Earth/Moon/Sun point-mass ephemeris-source replay, not full high-fidelity or flight validation. A scoped CasADi/IPOPT mature NLP backend bridge check for one normalized-CR3BP accepted case refines the polish nominal/branch-worst errors to 0.00914/0.01553 with IPOPT success 9/9 while preserving branch-recovery semantics. The hard-catalog tail-coast case is also normalized-CR3BP only: its all-configured one/two-segment row replays exactly from persisted accepted controls, but those hard-catalog controls fail bicircular solar-tidal stress and fixed-phase bicircular retuning. The contribution is therefore an auditable benchmark resource with calibrated negative and positive evidence, not fuel optimality, certified robustness, broad production-solver validation, or quantum advantage.

Keywords: Trajectory optimization benchmark, low-thrust trajectory optimization, cislunar dynamics, missed-thrust recovery, robust initialization, binary schedules, CR3BP, QUBO

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1. Introduction

Electric low-thrust propulsion enables propellant-efficient cislunar transfers, but it also makes trajectory design highly sensitive to the initial guess supplied to a nonlinear optimizer. A practical design must determine both continuous thrust magnitudes and directions and a qualitative maneuver structure: which intervals thrust, coast, or recover after an interruption. For astrodynamics researchers, this creates a reproducibility problem as much as an algorithm problem. A proposed initializer is difficult to interpret unless the schedule objective, missed-thrust masks, continuous recovery backend, evaluation budget, and failure cases are all visible.

Missed-thrust robustness is a demanding test of that pipeline. Safe modes, propulsion interruptions, or operations-driven thrust outages can invalidate a nominal low-thrust transfer. Grebow and McCarty show that including missed-thrust branches in low-thrust cislunar design can greatly reduce recovery propellant for week-scale outages [1]. Sinha and Beeson identify initial-guess generation as a bottleneck for low-thrust trajectory design with missed-thrust robustness [2]. This paper therefore asks a benchmark question rather than a quantum-performance question:

> Can a reproducible CR3BP benchmark resource reveal which missed-thrust initializers produce refinable schedules, which failure modes are caused by the continuous backend, and how surrogate-QUBO and simulated-QAOA initializers compare with strong classical samplers under equal trajectory-evaluation budgets?

Binary and QUBO formulations are included because they are a natural way to encode thrust-window availability, and because recent low-thrust studies have explored binary/QUBO and quantum-assisted formulations [3, 4]. They are not the headline claim. In this paper QUBO and QAOA are evaluated initializer families inside a broader robust-trajectory benchmark; the continuous trajectory remains a classical optimal-control problem, and negative evidence is reported as part of the contribution.

The paper makes four contributions for a trajectory-optimization audience. First, it defines and archives a normalized Earth–Moon CR3BP benchmark resource with source states, missed-thrust masks, configurations, result artifacts, generated tables/figures, metadata, and a machine-readable manifest. Second, it reports an equal-budget initializer comparison with a negative QAOA/sampler-superiority result: duty-cycle-aware schedules can be refinable, but the tested simulated QAOA variants do not dominate strong classical samplers or all-windows continuous refinement. Third, it adds a continuous-backend and replay/provenance ladder that separates selected-branch evidence, all-mask diagnostics, all-configured-mask evidence, accepted-control replay, and recorded force-model stress. Fourth, it records conservative external-validity semantics: cached Horizons force-model contrast, a positive simple bicircular phase-sweep stress replay, a cached-Horizons-derived solar-tidal replay, single- plus multi-epoch cached-Horizons Earth/Moon/Sun point-mass retuning packages, and a SPICE-derived ephemeris replay for the converged independent-HS all-configured row, plus negative bicircular solar-tidal stress and fixed-phase bicircular retuning outcomes for the hard-catalog tail-coast case.

The contribution is the schedule-initializer benchmark layer, not a replacement for production sequential-convex, collocation, or indirect low-thrust tools. Those mature solvers are downstream consumers and related-work comparators for a schedule or continuous-control initial guess. The controls used here–all-windows continuous refinement, direct shooting, compact collocation, continuation, and tail-coast diagnostics–are included to isolate whether the discrete schedule initializer adds value before handing the problem to stronger continuous machinery.

This framing is a natural fit for an astrodynamics methods journal even with negative quantum and robustness outcomes. The artifact is a cislunar trajectory-optimization resource: it exposes schedule-initializer failure modes, continuous-backend sensitivity, missed-thrust accounting, replay provenance, and force-model stress boundaries in a way that downstream astrodynamics solvers can use as controlled initialization tests. The value is therefore the audited benchmark and negative evidence, not a claim of a superior optimizer.

The claims are intentionally narrow. The experiments use normalized CR3BP dynamics, screening thresholds, configured outage families, and research-grade continuous backends. They do not establish quantum advantage, high-fidelity mission validation, fuel-optimal recovery, certified flight design, or universal missed-thrust robustness.

2. Related Work

2.1. Low-thrust trajectory optimization

Low-thrust trajectory optimization has a mature literature spanning indirect methods, direct transcription, collocation, global search, sequential convex programming, and hybrid approaches. Morante et al. survey low-thrust optimization as a hybrid optimal-control problem with both continuous and discrete structure [5]. In the Earth–Moon system, Ozimek and Howell computed low-thrust transfers to libration-point orbits and showed the role of CR3BP structure in preliminary design [6]. Pritchett, Howell, and Grebow used collocation, trajectory stacking, orbit chaining, and continuation to compute low-thrust transfers in the restricted three-body problem, including NRHO-to-DRO examples [7]. More recent cislunar and deep-space solvers include successive convex optimization for libration-point transfers [8], convex-programming approaches for rapid low-thrust trajectory design [9], thrust-regularized convex optimization [10], indirect forward-backward shooting in complex dynamics [11], and adaptive-mesh sequential convex programming [12]. These methods are mature trajectory-design baselines; the present work is not proposed as a replacement for them.

For preliminary cislunar design, the Circular Restricted Three-Body Problem is widely used because it captures the nonlinear Earth–Moon dynamical environment while keeping algorithmic studies tractable. The NASA/JPL Three-Body Periodic Orbits API provides normalized periodic-orbit states, periods, Jacobi constants, stability indices, and system constants for families such as halo orbits and distant retrograde orbits [13]. The catalog-derived benchmark in this paper uses those states as transparent boundary-condition seeds. Its purpose is to expose initialization and recovery failure modes, not to certify an operational transfer.

2.2. Robust and missed-thrust trajectory design

Missed-thrust design has been studied through margin analysis, expected-thrust formulations, virtual recovery trajectories, robust optimal control, and branching trajectory methods. Rubinsztejn et al. embed expected thrust fraction into deterministic trajectory design and report improved resilience to missed-thrust scenarios [14]. Venigalla et al. formulate multi-objective low-thrust trajectory optimization with missed-thrust recovery margin [15]. Grebow and McCarty specifically study low-thrust cislunar transfers, including NRHO-to-DRO transfers, and optimize branching trajectories for outage recovery [1]. Sinha and Beeson focus on initial-guess generation for robust low-thrust missed-thrust design and compare conditional and nonconditional global-search strategies [2]. Robust cislunar low-thrust optimization has also advanced through sequential convex programming and covariance steering; Kumagai and Oguri use chance-constrained covariance steering to design nominal trajectories and correction policies under uncertainty in the Earth–Moon CR3BP [16]. Broader uncertainty-aware low-thrust design includes nonlinear programming and forward-backward shooting under uncertainty [17], and missed-thrust robustness has recently been addressed with sequential convex programming [18]. State-dependent trust-region ideas for autonomous spacecraft SCP further illustrate the level of numerical machinery expected in robust flight-dynamics workflows [19]. These works set a high bar for any initializer: a useful discrete schedule must ultimately feed a continuous optimizer that can produce robust feasible trajectories.

2.3. QUBO/QAOA initializers

Quantum annealers and QAOA target discrete optimization models such as Ising Hamiltonians and QUBOs. A QUBO minimizes

$$E(x) = c + \sum_{i} a_i x_i + \sum_{i<j} b_{ij} x_i x_j, \qquad x_i \in \{0, 1\}, \tag{1}$$

which is convertible to the Ising form used in quantum annealing and many gate-model workflows [20, 21]. Farhi et al. introduced QAOA as a gate-model variational algorithm for approximate combinatorial optimization [22]. Shukla and Vedula formulated trajectory optimization as a discrete search problem and argued that quantum search can provide speedups for suitably discretized cases [23]. De Grossi et al. provide the closest low-thrust QUBO precedent [3]. This paper uses QUBO and QAOA only for the discrete initialization layer; the continuous trajectory remains a classical optimal-control problem.

2.4. Benchmark gap

The targeted catalog benchmark in this paper is deliberately harder than the teacher-controlled case because cislunar transfers between stable or nearly stable periodic orbits are known to be sensitive to the continuous initial guess. The gap addressed here is therefore diagnostic rather than algorithmic replacement: the paper asks how binary schedule samplers, including QUBO/QAOA variants, behave under equal trajectory-evaluation budgets when missed-thrust penalties are included and when their schedules are handed to research-grade continuous recovery backends. Existing missed-thrust and robust cislunar studies optimize trajectories, recovery branches, correction policies, or mission-design workflows. This paper instead packages reusable source states, masks, configurations, recorded controls, and evidence semantics so those stronger trajectory-design tools can be compared against a transparent initialization benchmark. Production SCP, collocation, and indirect tools remain the appropriate downstream solvers for mission-quality design. The all-windows continuous row and the continuous-backend diagnostics are the relevant controls for this benchmark layer because they test whether schedule selection improves the initial condition passed downstream. The resulting artifact package complements production-oriented studies by making failure modes, claim boundaries, and reproducibility provenance explicit.

3. Problem Formulation

3.1. Forced CR3BP dynamics

The spacecraft state is

$$s = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top \tag{2}$$

in the normalized Earth–Moon rotating frame. With mass parameter μ, distances to the primary and secondary bodies are

$$r_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \tag{3}$$

$$r_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}. \tag{4}$$

The forced CR3BP equations are

$$\ddot{x} = 2\dot{y} + x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3} + a_x, \tag{5}$$

$$\ddot{y} = -2\dot{x} + y - \frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3} + a_y, \tag{6}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3} + a_z. \tag{7}$$

The transfer duration T is divided into N uniform segments. A binary variable $x_i \in \{0,1\}$ gates thrust availability during segment i. During the discrete search stage, a deterministic target-seeking steering law maps the binary schedule to a propagated trajectory,

$$g(s) = K_r(r_f - r) + K_v(v_f - v), \qquad a_i(s) = x_i a_{\max} \frac{g(s)}{\|g(s)\| + \epsilon}. \tag{8}$$

This steering law is not claimed to be optimal. It creates a repeatable map from a bitstring to a trajectory so that discrete schedules can be ranked before continuous refinement. A separate oracle diagnostic, introduced only for the teacher-controlled benchmark, initializes refinement from the hidden continuous controls used to generate the target. That diagnostic is not a competing initializer; it tests whether the refinement formulation can recover the known feasible trajectory when continuous-control initialization is no longer the bottleneck.

3.2. Missed-thrust masks and robust binary cost

Let $\mathcal{M}$ denote a set of outage masks. Each mask zeros thrust in a one- or two-segment contiguous window. For a nominal schedule x, the outage-realized schedule is $x \odot m$. The scaled terminal error under mask m is

$$e_m(x) = \left\| \begin{bmatrix} (r_T(x \odot m) - r_f)/\ell_r \\ (v_T(x \odot m) - v_f)/\ell_v \end{bmatrix} \right\|_2. \tag{9}$$

The trajectory-evaluated robust binary objective is

$$\begin{aligned} J(x) = w_n e_0(x) + w_w \max_{m \in \mathcal{M}} e_m(x) + w_d \left(\max_{m \in \mathcal{M}} e_m(x) - e_0(x) \right) \\ + w_p \frac{1}{N} \sum_{i=1}^{N} x_i + w_s \frac{1}{N-1} \sum_{i=1}^{N-1} |x_{i+1} - x_i|. \end{aligned} \tag{10}$$

The terms penalize nominal miss distance, worst outage miss distance, missed-thrust degradation, active thrust fraction, and thrust/coast switching.

4. Initializer Families

4.1. QUBO surrogate

Because $J(x)$ depends on nonlinear propagation and outage evaluation, it is not itself a QUBO. The method therefore fits

$$\widehat{J}(x) = c + \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j \tag{11}$$

from trajectory-evaluated samples $\{x^{(k)}, J(x^{(k)})\}_{k=1}^{K}$ using ridge-regularized least squares:

$$\min_\theta \sum_{k=1}^{K} \left(J(x^{(k)}) - \widehat{J}(x^{(k)}; \theta) \right)^2 + \lambda \|\theta\|_2^2. \tag{12}$$

The fitted QUBO is then sampled by classical QUBO simulated annealing or converted to a diagonal cost Hamiltonian for statevector QAOA. The present experiments use QAOA only as a small simulated gate-model sampler; no hardware quantum advantage is implied.

4.2. Discrete initializers

The experiments compare random search, cross-entropy search, a genetic algorithm, true-objective simulated annealing, surrogate QUBO simulated annealing, and statevector QAOA. Random search and true-objective simulated annealing evaluate propagated schedules directly. Cross-entropy updates Bernoulli segment probabilities from elite samples. The genetic algorithm uses elitism, crossover, and mutation. The QUBO and QAOA methods use true trajectory evaluations to train $\widehat{J}$, then validate their best sampled schedules with the true objective.

Budget accounting is therefore reported in two ways: solver true evaluations exclude shared QUBO training, while total true evaluations include QUBO training. Equal total-evaluation comparisons are the relevant fair comparison.

4.3. Continuous branch-recovery refinement

The refinement stage converts a selected binary schedule into a continuous low-thrust candidate by optimizing piecewise-constant acceleration vectors only in active windows. Refined accelerations are projected to the Euclidean thrust ball $\|u_i\|_2 \leq a_{\max}$ before propagation and persistence. The branch-recovery mode optimizes a nominal control sequence plus selected-outage-specific post-outage recovery controls. Its least-squares residual includes nominal terminal error, selected outage terminal error, fuel regularization, control regularization, and inter-segment smoothness. A run is counted as successful only if both the nominal terminal error and the selected missed-thrust worst error are below the configured thresholds.

The paper also reports an all-mask diagnostic: after refinement on selected outages, every one- and two-segment outage mask is evaluated without claiming that all masks were optimized. This separates selected-branch recovery success from universal outage robustness.

5. Experimental Setup

5.1. Common dynamics and source state

All experiments use normalized Earth–Moon CR3BP dynamics with $\mu = 0.01215058560962404$. The common initial state is taken from the first row of an Earth–Moon L2 southern halo query with period in $[1.55, 1.65]$ TU from the NASA/JPL periodic-orbit API:

$$s_0 = [1.0324985738, 0, -0.1884844917, 0, -0.1249781287, 0]. \qquad (13)$$

This state is NRHO-like in the sense that it lies in the L2 southern halo family, but the paper treats it as a normalized benchmark state, not a certified operational NRHO.

5.2. Dimensional scale and threshold interpretation

The normalized Earth–Moon scale is used only to interpret orders of magnitude. Taking the public NASA average Earth–Moon distance as 1 LU $\approx$ 384,400 km [24], the recorded CR3BP normalization gives 1 TU $\approx$ 375,190.259 s = 4.34248 days, velocity unit $\approx$ 1.02455 km/s, and acceleration unit $\approx$ 2.73074 mm/s^2. Thus the main phase-shift transfer time $T = 0.5$ TU is about 2.17 days, while the hard-catalog tail-coast recovery case with $T = 4.0$ TU is about 17.37 days. The acceleration bounds used in the phase-shift and high-thrust diagnostics correspond to $a_{\max} = 0.2 \approx 0.546$ mm/s^2 and $a_{\max} = 1.0 \approx 2.731$ mm/s^2, respectively.

The terminal-error thresholds are screening tolerances in the normalized metric

$$\sqrt{\|\Delta r\|_2^2 + \|\Delta v\|_2^2/0.35^2}, \tag{14}$$

not flight navigation or operational targeting tolerances. If velocity error is zero, the nominal threshold 0.09 corresponds to a position-only error of about 34,596 km, and the selected-recovery threshold 0.17 corresponds to about 65,348 km. If position error is zero, the same thresholds correspond to velocity-only errors of about 32.3 m/s and 61.0 m/s under the configured velocity scale. This scale is the reason the manuscript frames the results as preliminary design and initializer benchmarking rather than operational targeting.

5.3. Benchmark families

Two benchmark families define the main evidence path. The harder catalog-derived family uses a distant retrograde orbit seed from an Earth–Moon DRO query,

$$s_{\mathrm{DRO}} = [0.8161260827, 0, 0, 0, 0.5076556931, 0], \tag{15}$$

and generates the target by ballistic propagation of that seed. The original catalog-derived schedule run uses $T = 2.7$ TU, $N = 12$, $a_{\max} = 0.055$, and one- and two-segment outage masks. It remains negative evidence in the supplement. The main hard-catalog result instead reports a later locked-nominal tail-coast continuous-backend diagnostic at $T = 4.0$ TU, $N = 14$, and $a_{\max} = 1.0$, where the final five nominal control segments are constrained to zero and all configured one- and two-segment masks are selected and evaluated in one fixed-final-time row.

The non-teacher phase-shift family is derived from the same JPL L2 southern halo source state s_0. The target is generated by zero-thrust CR3BP propagation of s_0 for a phase time of 0.3 TU, while the transfer must arrive at that target after $T = 0.5$ TU. A zero-thrust trajectory has normalized terminal error 0.3758, so the case is not a do-nothing pass. The primary initializer benchmark uses $N = 12$, $a_{\max} = 0.2$, one-segment outage masks, and a duty-cycle prior

$$J_\rho(x) = w_\rho \left(\frac{1}{N} \sum_{i=1}^{N} x_i - \rho \right)^2, \tag{16}$$

with $\rho = 11/12$ and $w_\rho = 12$. This prior is not an oracle trajectory correction; it encodes a preference for one coast window while preserving a dense low-thrust arc.

The supplement preserves the secondary diagnostic families: the catalog pilot and QUBO fit diagnostics, feasibility and direct multiple-shooting stress tests, locked-nominal and delayed-arrival hard-catalog recovery packages, bounded phase-suite and robust-margin rows, direct-collocation and Hermite–Simpson baselines, dense-repair and legacy cardinality ablations, and the teacher-controlled feasible benchmark. These diagnostics explain provenance and failure modes but are not promoted to headline evidence in the main article.

5.4. Metrics

The primary metrics are refined nominal terminal error, refined selected missed-thrust worst error, all-mask diagnostic worst error, schedule active fraction, nominal and recovery fuel, total true trajectory evaluations, and refinement success rate. A run is counted as a robust refinement success only when both the nominal terminal error and the selected missed-thrust worst error are below the configured thresholds. Unless otherwise stated, those thresholds are 0.09 for nominal success and 0.17 for selected robust recovery success.

The main $N = 12$ duty-cycle-prior statistics package repeats seven methods over 30 deterministic seeds, for 210 raw method–seed rows, and reports Wilson success intervals, paired per-seed success differences, bootstrap confidence intervals for mean selected-worst-error differences, and exact two-sided sign tests. A separate focused QAOA-depth package repeats the QUBO-derived methods over 30 deterministic seeds, for 150 raw method–seed rows. The hard-catalog tail-coast row `tail_coast_all_one_two_segment_t5_portfolio` uses the same thresholds, selects all configured one- and two-segment masks in one case row, and reports optimizer convergence separately from threshold feasibility so late-tail direct evaluations without recovery variables are not mistaken for optimizer-converged branch solves.

6. Results

The main results are screening evidence for a normalized CR3BP benchmark. The dimensional caveat in Section 5.2 applies to every table: the 0.09/0.17 thresholds correspond to large position-only or velocity-only errors and should not be read as operational targeting tolerances. Secondary tables and figures are retained in the supplement so that the main article follows the primary claim path, while the evidence-synthesis table below cross-indexes representative rows from those recorded artifacts.

6.1. Claim hierarchy

Table 1 records the claim boundary used throughout the paper. It lists each headline claim, its primary artifacts, its status, and interpretations that are explicitly prohibited by the evidence.

6.2. Claim-ledger summary

The generated claim ledger in `data/results/claim_evidence_ledger/` is retained as a reviewer-facing audit artifact rather than as a dense main-text table. It reads recorded CSV/JSON files only and separates selected-branch evidence, all-mask diagnostics, all-configured-mask rows, accepted-control replay, force-model contrast, stress probes, and retuned-recovery evidence. The full ledger, tail-coast threshold audit, and tail-coast branch audit are reported in the supplement. The main text keeps the compact hierarchy in Table 1 and the cross-backend synthesis in Table 2; together they state the operative boundaries without turning the article into an artifact listing.

For practitioners, the reporting lesson is concrete: include the all-windows continuous control, do not collapse selected-branch and all-mask semantics, preserve negative QAOA/QUBO evidence, and archive provenance in machine-readable artifacts. In the current ledger, the new independent-midpoint Hermite–Simpson headroom package is all-configured one-segment normalized-CR3BP evidence with separate branch-control replay, simple bicircular phase-stress, cached-Horizons-derived solar-tidal replay, single- plus multi-epoch cached-Horizons point-mass retuning rows, and a SPICE-derived replay row, while the earlier independent-HS p=0.4 baseline row remains selected-branch/all-mask diagnostic evidence.

6.3. Cross-backend evidence synthesis

Table 2 is generated by `scripts/run_evidence_synthesis.py`. The script reads only recorded CSV/JSON artifacts and writes `data/results/evidence_synthesis/evidence_synthesis.csv`, `data/results/evidence_synthesis/evidence_synthesis_metadata.json`, and the LaTeX table used here. It is not a new optimization run. Its purpose is to put the broader validation axes into the main claim path without overstating them.

The synthesis clarifies the positive contribution. The 30-seed threshold-sensitivity row shows that at the tight $(0.050, 0.090)$ check every sampled method is 0/30 while all-windows continuous remains 30/30, so the paper does not claim sampler superiority. The continuation-extension rows show that all-mask direct multiple-shooting continuation can meet a tighter $(0.065, 0.100)$ phase-shift screen for all one-segment masks and for all configured one/two masks, but not the 0.05 nominal headroom check. The direct-collocation and earlier independent-midpoint Hermite–Simpson rows show compact continuous backends with tight selected/all diagnostics while preserving selected-branch semantics. The new independent-midpoint Hermite–Simpson headroom package upgrades that phase-shift evidence to all-configured one-segment rows at phase time 0.4 and $a_{\max} = 0.2$: all eight configured masks are selected and evaluated, with the original row at nominal error 0.0111 and selected/all worst error 0.0774, and a polished converged row at nominal error 0.0113 and selected/all worst error 0.0779. A separate simple bicircular phase-sweep stress replay validates the persisted polish-row controls over eight Sun phases: all 8 nominal phases and all 64 branch-phase checks pass, with maximum nominal error 0.022138676654057693 and maximum branch-phase error 0.08557051343145317. A cached-Horizons-derived solar-tidal replay for the same polish row uses a representative 2026-Jan-01 JPL Horizons geometry cache and also passes, with nominal error 0.018363195236986728, branch worst 0.07422350563850917, and branch pass count 8/8. A stronger cached-Horizons Earth/Moon/Sun point-mass model at the same representative epoch records a negative direct replay for persisted controls, with nominal error 0.3812580376880591 and branch worst 0.3797450961017463, but independent least-squares retuning restores feasibility with retuned nominal error 0.02143944130524006, retuned branch worst 0.02473065115224942, and branch pass count 8/8. The four-epoch multi-epoch retuning wrapper keeps the retuned branch pass count at 32/32, and the SPICE-derived replay of those 36 retuned controls keeps nominal/branch worst errors at 0.021439441253166033/0.024730650824609506 with maximum delta $3.2763991519857427 \times 10^{-10}$ relative to the Horizons-retuned replay. The hard-catalog rows show that the fixed-final-time pass is scoped to the tail-coast locked-nominal backend; an independent-HS selected-outage hard-catalog diagnostic remains a negative result.

6.4. Tail-coast hard-catalog diagnostic

Table 3 reports the strongest hard-catalog fixed-final-time result. The setup solves and freezes a locked nominal control, then refines it with the final five nominal control segments constrained to exact zero. Branches keep the original target and original final time; no delayed target or added recovery horizon is used. The nominal-only provenance row has tail-coast nominal error 0.022992 and therefore passes the nominal screening threshold, while its masked-nominal all-mask diagnostic remains 27.8351 because no branch recovery is run.

The headline row is `tail_coast_all_one_two_segment_t5_portfolio`. It sets `outage_lengths=[1,2]`, selects all configured masks, and evaluates all 27 selected masks in one fixed-final-time case row. The row reaches tail-coast nominal error 0.022992338178558882, selected fixed-time worst error 0.0936063931709301, and all-mask fixed-time diagnostic worst error 0.0936063931709301, so it meets the configured 0.09/0.17 screening thresholds. The branch portfolio evaluates deterministic fallback starts for four branches, charges 5929 function evaluations, and records runtime about 1833.6 s in the current CSV. Among branches with post-outage recovery variables, the optimizer-converged count is 25/25. The remaining two late-tail masks have no post-outage recovery variables; they are direct fixed-final-time threshold evaluations rather than optimizer-converged branch solves. Therefore `branch_optimizer_all_success=false` remains a conservative CSV accounting flag even though `meets_thresholds=true`.

This result is deliberately scoped. It is fixed-final-time continuous-backend evidence with a locked nominal control, final-five-segment coast, configured one- and two-segment masks, fallback starts, and late-tail no-recovery-variable accounting. It is not a binary-sampler result, not QUBO/QAOA evidence, not fuel optimality, not high-fidelity validation, and not robustness to outage families beyond the configured masks.

6.5. Replay and external-validity boundary

The full accepted-control replay, nominal sidecar replay/stress, independent-HS branch-control replay, independent-HS cached-Horizons solar-tidal replay, independent-HS cached-Horizons point-mass retuning, independent-HS SPICE-derived ephemeris replay, Horizons contrast, bicircular stress, and bicircular retuning tables are moved to the supplement. The main facts are concise. The independent-HS branch-control replay contains two p=0.4 nominal rows and 16 branch rows; repropagation of persisted endpoint and midpoint schedules under the configured normalized CR3BP model gives zero nominal, branch, selected-worst, and all-mask-worst terminal-error deltas at tolerance 10^{-10}. The focused accepted-control replay for `tail_coast_all_one_two_segment_t5_portfolio` contains one nominal row and 27 branch rows; repropagation under the configured normalized CR3BP model gives zero nominal, branch, and maximum terminal-error delta at tolerance 10^{-10}. Both replay packages are deterministic replay of persisted controls, not new solves or mature-backend portability/bridge checks.

The external-validity checks are deliberately mixed and conservative. A cached JPL Horizons contrast for the same hard-catalog window samples 15 transfer nodes from a fixed 2026-Jan-01 TDB epoch and records Earth–Moon distance/rate variation, Sun distance/phase differences, and solar-tidal acceleration differences; it is a force-model contrast only, not SPICE validation or ephemeris propagation. The converged independent-HS all-configured polish row provides a positive deterministic stress-probe result under the simple circular solar-tidal model: the persisted endpoint-plus-midpoint controls pass all 8 nominal Sun phases and all 64 branch-phase checks with maximum errors 0.022138676654057693 and 0.08557051343145317. The same polish row also passes a cached-Horizons-derived solar-tidal replay at the representative 2026-Jan-01 epoch, with nominal error 0.018363195236986728, branch worst 0.07422350563850917, branch pass count 8/8, CR3BP replay delta 0.0, and Sun distance range 382.6857920288508–382.693044178952 LU. This uses cached JPL Horizons geometry but remains a simplified stress probe, not SPICE/high-fidelity/flight validation. Under a stronger geocentric Earth/Moon/Sun point-mass model using the same cache, direct replay of the persisted polish controls fails with nominal error 0.3812580376880591 and branch worst error 0.3797450961017463. Independent least-squares retuning of the nominal and branch endpoint-plus-midpoint controls restores feasibility at that representative epoch, with retuned nominal error 0.02143944130524006, retuned branch worst 0.02473065115224942, branch pass count 8/8, optimizer success count 9/9, and cache SHA-256 prefix `13fe6993`; the full hash is retained in the supplement and metadata. The four-epoch cached-Horizons wrapper repeats that point-mass retuning protocol at Jan/Apr/Jul/Oct 2026, and the SPICE-derived replay then repropagates the 36 already-retuned controls with compact Moon/Sun vector caches derived from `naif0012.tls`, `de442s.bsp`, and `gm_de440.tpc`: all 4 nominal rows and all 32 branch rows pass, worst nominal/branch errors are 0.021439441253166033/0.024730650824609506, and the maximum delta from the Horizons-retuned replay is $3.2763991519857427 \times 10^{-10}$. This is a point-mass ephemeris-source replay package, not full high-fidelity or flight validation, broad production-solver validation/parity, fuel optimality, or quantum evidence. In contrast, replaying the hard-catalog accepted controls with a simple bicircular solar-tidal term fails the configured thresholds: every nominal solar-tidal phase fails the 0.09 nominal threshold, only 22/108 branch-phase rows pass the 0.17 branch threshold, and the maximum solar-tidal terminal error is 37.7596. A fixed-phase retuning batch under the same simple bicircular model also remains negative: retuned nominal error is 0.316772, configured branch pass count is 19/27, maximum retuned branch error is 6.0299, and strict branch pass count is 16/27. The hard-catalog tail-coast pass is therefore foregrounded as normalized-CR3BP evidence only, while the independent-HS stress, retuning, and ephemeris-source replay results are representative-epoch probes rather than high-fidelity validation.

6.6. Thirty-seed main-method statistics

Tables 4 and 5, together with Figure 1, repeat the configured $N = 12$ duty-cycle-prior method comparison over 30 deterministic seeds. The raw package contains 210 rows, i.e., 30 seeds times seven methods: random search, cross-entropy search, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, the configured statevector QAOA sampler, and all-windows continuous refinement.

The success-rate evidence strengthens the 10-seed result but does not create a quantum-performance claim. Random search succeeds in 11/30 seeds, with Wilson 95% interval $[0.219, 0.545]$. Cross-entropy, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement each succeed in 30/30 seeds, with Wilson interval $[0.886, 1.000]$. The configured `qaoa_statevector` method succeeds in 13/30 seeds, with interval $[0.274, 0.608]$.

The paired error comparisons are more important for the aerospace interpretation. Against all-windows continuous refinement, every sampled method has a positive mean selected-worst-error difference; negative would favor the sampled method. The exact two-sided sign-test value is approximately 1.86×10^{-9} for each sampler versus all-windows continuous refinement. Thus, the all-windows continuous baseline is not just equally successful for the strongest methods; it also gives lower selected-worst error on this benchmark. Against surrogate-QUBO simulated annealing, cross-entropy, genetic search, and true-objective simulated annealing tie the success rate and have small nonsignificant selected-worst-error deltas. Random search is worse. The configured statevector QAOA row has lower success and a positive mean selected-worst-error delta of 0.0153 versus surrogate-QUBO simulated annealing, with exact sign-test $p = 0.1849$.

This package therefore strengthens the statistics beyond a QAOA-only ablation while preserving the conservative conclusion. Duty-cycle priors help several samplers find feasible high-duty schedules, but all-windows continuous fuel regularization remains a stronger mechanism for selected-worst error in this benchmark, and the evidence does not support quantum advantage or QAOA superiority.

Table 6 adds a threshold-sensitivity sanity check computed only from `data/results/phase_shift_cardinality_30seed/raw_results.csv`. At the stricter $(0.075, 0.120)$ nominal/selected-worst thresholds, cross-entropy, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement still pass 30/30 seeds, while random search falls to 4/30 and `qaoa_statevector` remains 13/30. At $(0.065, 0.100)$ the sampled high-duty schedules largely stop passing, but all-windows continuous remains 30/30. At the tight $(0.050, 0.090)$ check, every

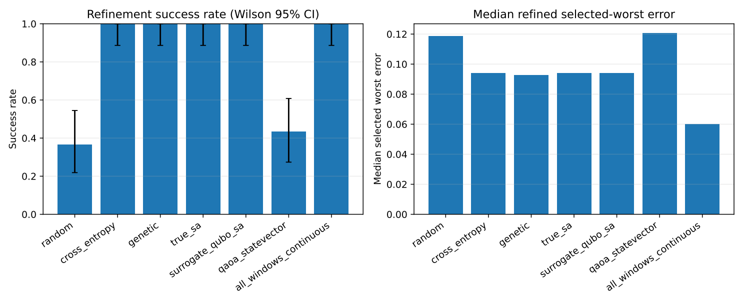

Figure 1: Thirty-seed main-method statistics for the duty-cycle-prior benchmark. High-duty samplers can be feasible, but all-windows continuous refinement remains lower-error on paired selected-worst comparisons.

sampled method is 0/30 while all-windows continuous is still 30/30. Thus stricter thresholds do not rescue sampled-method superiority; they quantify the margin limitation of the sampled schedules and reinforce the all-windows continuous lower-error diagnostic.

6.7. Focused QAOA/QUBO ablation

Table 7 and Figure 2 isolate the QUBO-sampling step on the same $N = 12$ duty-cycle-prior benchmark with 30 deterministic seeds. Each QUBO-derived method uses the same accounting per seed: 96 shared trajectory evaluations to train the QUBO and 24 true post-QUBO candidate evaluations. The resulting raw package contains 150 method–seed rows; the raw method summary table is in the supplement.

The 30-seed success rates with Wilson 95% intervals are $28/30 = 0.933$ $[0.787, 0.982]$ for surrogate-QUBO simulated annealing, $28/30 = 0.933$ $[0.787, 0.982]$ for exact evaluation of the top 24 QUBO bitstrings, $15/30 = 0.500$ $[0.332, 0.668]$ for random-angle $p = 1$ QAOA, $21/30 = 0.700$ $[0.521, 0.833]$ for optimized $p = 1$ QAOA, and $29/30 = 0.967$ $[0.833, 0.994]$ for optimized $p = 2$ QAOA. Optimized $p = 2$ is competitive with the strongest classical QUBO sampler in this narrow statevector QUBO ablation, but it is not statistically supported as superior: versus surrogate-QUBO simulated annealing, its paired success delta is only $+0.033$, the mean selected-worst-error delta is -0.00056 with bootstrap 95% confidence interval $[-0.00393, 0.00236]$, and the exact sign-test p-value is 0.4545.

The useful positive result is narrower. Relative to random-angle $p = 1$ QAOA, optimized $p = 2$ QAOA increases success by $+0.467$ and

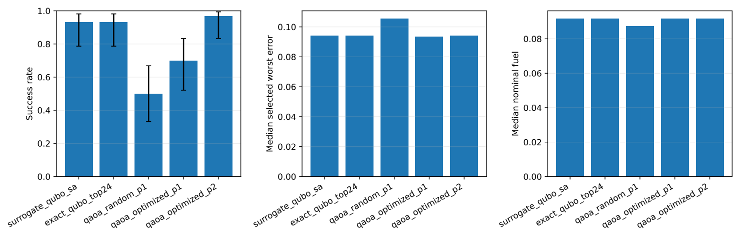

Figure 2: Thirty-seed QAOA-depth ablation summary. Optimized $p = 2$ QAOA improves over random-angle QAOA and is competitive with classical QUBO sampling, but paired tests do not support a superiority claim over surrogate-QUBO simulated annealing.

reduces the mean selected-worst error by 0.0125 with bootstrap interval $[-0.0188, -0.0060]$. The exact sign-test p-value is still not significant at conventional levels ($p = 0.2478$), so the evidence should be read as practical improvement in this ablation rather than a formal dominance claim. No quantum advantage is claimed.

6.8. Supplementary diagnostics

The supplement preserves the evidence that explains how the headline results were reached and where they fail. This includes the original catalog pilot and QUBO fit, feasibility sweeps, direct multiple-shooting and targeted catalog attempts, selected-outage hard-catalog envelope, locked-nominal and delayed-arrival hard-catalog recovery packages, bounded phase-suite and robust-margin rows, direct-collocation and Hermite–Simpson baselines, unrepaired and repaired $N = 8$ phase-shift evidence, legacy 10-seed cardinality-prior summaries, cardinality and fuel ablations, and the teacher-controlled benchmark. These artifacts support the failure-mode discussion, but the main claims do not depend on promoting them to primary results.

7. Discussion

The central result is a benchmark result, not a sampler-performance claim. Binary thrust-window schedules can be embedded in a reproducible missed-thrust initialization workflow, but the usefulness of a schedule is controlled by the continuous recovery basin. The supplementary diagnostics show both sides: sparse binary schedules can fail even when all-windows continuous refinement succeeds, while duty-cycle priors can preserve enough thrust availability for several samplers to find refinable high-duty schedules.

The 30-seed main-method package is the most relevant initializer evidence. Cross-entropy search, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement all succeed in 30/30 seeds on the duty-cycle-prior benchmark, whereas the configured statevector QAOA row succeeds in 13/30 seeds. The paired selected-worst-error tests favor all-windows continuous refinement over every sampled method, and the stricter-threshold sensitivity check shows the sampled high-duty schedules losing pass counts before all-windows continuous does. Thus the paper supports duty-cycle-aware robust initialization as a useful benchmark mechanism, but not a claim that a binary sampler improves the continuous trajectory beyond a strong dense-control baseline.

The focused QAOA/QUBO ablation provides a more favorable but still bounded view of QAOA. Optimizing the angles and increasing simulated depth to $p = 2$ improves QAOA relative to random-angle $p = 1$ QAOA and is competitive with classical QUBO sampling in that narrow statevector study. The paired comparisons do not support superiority over surrogate-QUBO simulated annealing, and the result is not hardware evidence. For this benchmark, QAOA is best interpreted as a measured initializer family with negative and inconclusive comparisons, not as the contribution itself.

The continuous-backend ladder is useful because it separates what was optimized from what was only diagnosed. The earlier independent-midpoint Hermite–Simpson p=0.4, $a_{\max} = 0.2$ row had strong selected-branch and all-mask diagnostic errors, but it optimized only three selected outage branches. The new all-configured headroom package closes that semantic gap for the one-segment phase-shift family: all eight configured masks are optimized and evaluated. The original p=0.4 row records nominal error 0.0111 and selected/all worst error 0.0774 while reaching its configured function-evaluation cap; the polished row warm-started from the original row and its branch controls, converges, and records nominal error 0.0113 and selected/all worst error 0.0779. The replay/stress package repropagates nominal endpoint and midpoint controls at the source integration grid, refined substep grids, and direct $\pm 1\%$ acceleration scales. The new branch-control replay package separately repropagates the persisted full endpoint and midpoint branch schedules for both p=0.4 rows and records zero deltas for 16 branch rows. The bicircular phase-sweep stress package then reuses those same persisted endpoint-plus-midpoint controls under a simple circular solar-tidal perturbation and records a positive polish-row stress result over eight phases, with all nominal and branch-phase checks below the configured thresholds. The cached-Horizons-derived replay uses the same controls with interpolated Sun vectors from a representative 2026-Jan-01 JPL Horizons cache and records nominal error 0.018363195236986728 and branch worst 0.07422350563850917. The cached-Horizons Earth/Moon/Sun point-mass package records the harder negative direct replay first, then reruns independent retuning and restores retuned nominal/branch-worst errors 0.02143944130524006/0.02473065115224942 with branch pass count 8/8; the multi-epoch wrapper repeats the same point-mass retuning protocol at four fixed 2026 epochs, shows nominal direct replay fails in each epoch, and retains direct branch pass count 18/32 overall (July 8/8) plus worst retuned nominal/branch errors 0.02143944130524006/0.02473065115224942 with retuned branch pass count 32/32. The SPICE-derived replay is the next rung: it does not retune, but replays those 36 retuned controls against compact SPICE-derived Moon/Sun vector caches and records nominal/branch worst errors 0.021439441253166033/0.024730650824609506 with maximum Horizons-to-SPICE replay delta $3.2763991519857427 \times 10^{-10}$. Finally, the CasADi/IPOPT bridge exports the accepted normalized-CR3BP polish nominal and eight branch sidecars to a mature NLP backend, fixes each branch pre-recovery prefix to the refined nominal masked controls, and locally refines the source replay errors 0.011333095366088189/0.07792080291839382 to 0.009138565365046585/0.015534969964216154 with IPOPT success 9/9 and zero recorded prefix and control-bound violations. This is normalized-CR3BP continuous-backend evidence plus deterministic beyond-CR3BP stress, retuning, ephemeris-source replay, and a scoped mature-backend portability/bridge check with explicit optimizer-status and model-scope caveats, not production mission design or high-fidelity validation.

The hard-catalog tail-coast diagnostic addresses a different reviewer concern: whether any hard-catalog missed-thrust recovery case can meet the configured fixed-final-time thresholds once nominal feasibility is locked. The combined one/two-segment row does meet those thresholds, but only within a carefully scoped continuous-backend setup. The final-five-segment coast, locked nominal control, fallback starts, 25/25 optimizer-converged eligible recovery branches, and two no-recovery-variable late-tail direct evaluations are part of the result. The accepted controls replay exactly under normalized CR3BP but fail the bicircular solar-tidal stress sweep without retuning. This contrasts with the positive independent-HS simple bicircular and cached-Horizons-derived stress replays and keeps the hard-catalog result demoted to a stress limitation. A completed fixed-phase retuning batch under the same simple bicircular solar-tidal model also remains negative: at Sun phase 0°, with the original fixed target and final time and all 27 one/two-segment masks, the retuned nominal error is 0.316772, configured branch pass count is 19/27, maximum retuned branch error is 6.0299, and the strict $(0.05, 0.09)$ branch pass count is 16/27. This narrows a backend blocker and records useful recovery provenance, but it does not establish fuel-optimality, high-fidelity robustness, broader outage-family robustness, or quantum evidence.

The practical implication for astrodynamics researchers is that binary schedule benchmarks should report continuous-backend diagnostics and negative evidence as first-class artifacts. A stronger future initializer claim would need better trajectory-shaped binary encodings, mature continuous optimal-control baselines such as production collocation or sequential convex programming workflows, high-fidelity validation, and statistically significant improvements over strong classical initializers under equal total trajectory-evaluation budgets.

8. Limitations

The normalized CR3BP experiments are preliminary-design algorithm studies. They omit ephemeris dynamics, mass depletion, eclipse constraints, navigation uncertainty, thrust pointing limits beyond the acceleration norm, operational constraints, and high-fidelity force modeling. The terminal-error thresholds are screening tolerances; their dimensional interpretation is deliberately large and not comparable to flight targeting requirements.

The continuous backends are research-grade direct shooting, compact direct collocation, Hermite–Simpson, and continuation diagnostics. They are useful for comparing initializer behavior and exposing failure modes, but they are not certified flight-design tools. The new independent-midpoint Hermite–Simpson all-configured package is restricted to the normalized CR3BP one-segment phase-shift family: the original p=0.4 row reaches its configured function-evaluation cap, while the polished p=0.4 row converges with slightly larger recorded errors. The recorded-control replay/stress table checks deterministic nominal sidecar replay only. The independent-HS branch-control replay and the focused tail-coast accepted-control replay check persisted branch-control sidecars under normalized CR3BP only; they are not optimization reruns, high-fidelity validation, broad production-solver validation/parity, fuel-optimal analysis, or quantum evidence. The independent-HS bicircular phase-sweep replay is a positive deterministic stress probe under a simple circular solar-tidal model, not SPICE/high-fidelity validation or broad production-solver validation/parity. The independent-HS cached-Horizons replay uses cached JPL Horizons Moon/Sun geometry for a representative 2026-Jan-01 simplified solar-tidal replay, but it is not SPICE propagation, high-fidelity/flight validation, flight-ready evidence, or broad production-solver validation/parity. The independent-HS cached-Horizons point-mass packages rerun local least-squares retuning after documenting persisted-control replay failure; the new multi-epoch wrapper addresses the single-epoch concern over four fixed 2026 cached-Horizons epochs, but both packages remain point-mass stress/retuning results, not SPICE propagation, full high-fidelity or flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence. The independent-HS SPICE-derived replay reuses the already-retuned controls against compact SPICE-derived Moon/Sun vector caches without retuning or optimization rerun; it is an Earth/Moon/Sun point-mass ephemeris-source replay, not full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence. The independent-HS CasADi/IPOPT bridge reruns a local direct-shooting refinement in a mature NLP backend for one accepted normalized-CR3BP polish case while fixing branch pre-recovery prefixes to refined nominal masked controls; it is a scoped CasADi/IPOPT mature NLP backend bridge check, not production mission design, high-fidelity or flight validation, global optimality, fuel optimality, DOI evidence, or quantum evidence. The Horizons ephemeris package is an offline force-model contrast against cached Earth/Moon/Sun geometry, not SPICE validation or ephemeris trajectory replay. The hard-catalog bicircular solar-tidal replay is a simple beyond-CR3BP stress probe and its threshold failures are an external-validity limitation, not SPICE validation. The bicircular retuned recovery package is likewise a simple fixed-phase retuning experiment under the same circular solar-tidal model; even after retuning all configured masks it fails the recorded thresholds and is not SPICE/high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, or quantum/QUBO/QAOA evidence. The claim evidence ledger clarifies selected-branch, all-mask diagnostic, all-configured-mask, branch-control replay, force-model-contrast, stress-probe, retuned-recovery, SPICE-derived ephemeris-replay, and CasADi/IPOPT bridge semantics, but it does not add high-fidelity validation. Many supplementary rows optimize selected outage branches and report unselected outage masks diagnostically; only rows that explicitly select all configured masks should be read as all-configured-mask evidence.

The hard-catalog tail-coast result is scoped to a locked-nominal continuous-backend setup with final-five-segment coast, configured one- and two-segment masks, deterministic fallback starts, and late-tail no-recovery-variable direct evaluations. The same controls do not pass the configured bicircular solar-tidal stress sweep, the retuned fixed-phase bicircular batch still fails after retuning, and the cached Horizons contrast shows distance, angular-rate, phase, and solar-tidal acceleration differences not represented by the simple circular stress model. The positive independent-HS simple bicircular phase-sweep and cached-Horizons-derived solar-tidal replays, plus the independent-HS cached-Horizons point-mass retuning packages and SPICE-derived replay, are separate one-segment phase-shift stress/retuning/replay results and do not upgrade the hard-catalog row to beyond-CR3BP robustness. The hard-catalog result therefore does not prove fuel-optimal recovery, high-fidelity robustness, certified flight design, or broader outage-family robustness.

The discrete-initializer conclusions are also bounded. The 30-seed packages improve statistical rigor, but QAOA is simulated with shallow depths and limited optimizer budgets, and the evidence does not support quantum advantage or QAOA superiority. Smaller supplementary packages, including teacher-controlled and legacy 10-seed results, are retained as diagnostics rather than as strong stochastic rankings.

9. Reproducibility

The repository contains Python source, configurations, source-state metadata, generated tables, figures, result metadata, cached Horizons and compact SPICE-derived vector data, PDFs, and a scoped artifact manifest. The supplement and `REPRODUCIBILITY.md` provide the full artifact map, including the 30-seed initializer packages, claim-evidence ledger, evidence-synthesis tables, independent-midpoint Hermite–Simpson all-configured headroom package, replay/stress sidecars, independent-HS branch-control replay, independent-HS simple bicircular phase-stress replay, independent-HS cached-Horizons solar-tidal replay, single- and multi-epoch independent-HS cached-Horizons point-mass retuning, independent-HS SPICE-derived ephemeris replay, independent-HS CasADi/IPOPT bridge artifacts, accepted tail-coast branch-control replay, cached Horizons contrast, hard-catalog bicircular stress, and negative bicircular retuning artifacts.

The short reviewer-facing check is the read-only command `py -3.11 scripts/verify_submission_snapshot.py`. It checks key primary artifact paths, runs `scripts/write_artifact_manifest.py` with `-check`, runs the focused reproducibility pytest subset, and runs `git diff -check` unless disabled by a flag. It does not regenerate artifacts, rerun trajectory optimization, rebuild PDFs, or remove LaTeX auxiliary files; `-full-tests` expands the pytest step to the full suite. Evidence-regeneration commands and long-run caveats remain in `REPRODUCIBILITY.md`.

The manifest is generated by `scripts/write_artifact_manifest.py`. Its scoped SHA-256 hashes and byte counts are the authoritative artifact identities. The field `git_head_at_generation` records the repository state when the manifest was produced; for a committed manifest this necessarily refers to the parent/pre-final state rather than the final commit containing the refreshed manifest. Historical long-run metadata may predate the repository or contain dirty-state records. The current submission snapshot is verified by clean git status after final commit, manifest checking, tests, and local LaTeX builds rather than by interpreting old experiment metadata as final provenance.

10. Conclusion

This paper contributes a reproducible normalized CR3BP benchmark resource for robust low-thrust cislunar initialization under missed-thrust events. Its main finding is conservative: duty-cycle-aware binary schedules can help locate refinable high-duty candidates, but all-windows continuous refinement remains lower-error in paired selected-worst comparisons and remains more robust under stricter threshold checks, while the tested simulated QAOA variants do not establish superiority over classical initializers. Deterministic evidence-ledger, evidence-synthesis, sidecar-replay, cached Horizons force-model contrast, and solar-tidal stress layers make the broader validation axes explicit: all-mask continuation, compact collocation, and the new all-configured independent-midpoint Hermite–Simpson package provide phase-shift CR3BP headroom; recorded nominal controls replay consistently; persisted independent-HS branch controls replay exactly under normalized CR3BP; the converged independent-HS polish row passes the simple bicircular phase-sweep stress replay with maximum nominal/branch errors 0.02214/0.08557 and a cached-Horizons-derived representative-epoch replay with errors 0.01836/0.07422; the stronger cached-Horizons point-mass model fails direct replay but independent retuning restores nominal/branch-worst errors 0.02144/0.02473 for all eight branches at the January epoch and across the four-epoch wrapper with branch pass count 32/32; the SPICE-derived replay of those retuned controls preserves branch pass count 32/32 with maximum Horizons-to-SPICE replay delta 3.28×10^{-10}; a scoped CasADi/IPOPT bridge exports the accepted polish controls to a mature NLP backend and refines normalized-CR3BP nominal/branch-worst errors to 0.00914/0.01553 while preserving branch-recovery semantics with IPOPT success 9/9; the hard-catalog pass still requires the scoped tail-coast backend; the accepted hard-catalog branch-control sidecars replay exactly under the configured normalized CR3BP model; the cached ephemeris contrast quantifies simple circular force-model omissions; and the hard-catalog controls fail the configured bicircular solar-tidal stress sweep. A completed simple bicircular retuning batch at fixed Sun phase 0° also fails rather than upgrading the hard-catalog result to beyond-CR3BP validation. The useful outcome is therefore an auditable benchmark with negative evidence, failure modes, and provenance, not a claim of quantum advantage or flight-ready robustness.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

Code, generated artifacts, and artifact maps for the current draft are available in the working repository associated with this revision. The `v1.0.0` tagged repository/artifact snapshot at commit `1164cb36367ca37d4860e7f137cba2afc67b2999` is externally archived at Zenodo with DOI 10.5281/zenodo.20739420. This post-deposit repository revision records that DOI metadata; the DOI-bearing metadata edits themselves are not part of the `v1.0.0` Zenodo source archive. The benchmark source states are derived from the NASA/JPL Three-Body Periodic Orbits API and recorded in `data/source_states.json`. The Horizons force-model contrast, independent-HS cached-Horizons solar-tidal replay, and single- and multi-epoch independent-HS cached-Horizons point-mass retuning packages run offline from committed caches in `data/cache/horizons/`, whose metadata records exact API URLs and query parameters. The SPICE-derived replay runs offline from committed compact vector caches in `data/cache/spice/`; refreshing those caches downloads NAIF kernels locally, records their SHA-256 values, and does not archive raw kernel binaries. The CasADi/IPOPT bridge runs offline from committed independent-HS sidecars with `casadi==3.7.2`. `ARCHIVAL_RELEASE.md` records the external Zenodo DOI metadata for the archive snapshot.

References

[1] D. J. Grebow, S. L. McCarty, Missed thrust analysis and design for low thrust cislunar transfers, Tech. Rep. NTRS 20220001503, Jet Propulsion Laboratory, National Aeronautics and Space Administration (2020).
URL https://ntrs.nasa.gov/citations/20220001503

[2] A. Sinha, R. Beeson, Initial guess generation for low-thrust trajectory design with robustness to missed-thrust events, Journal of Guidance, Control, and Dynamics 48 (12) (2025) 2831–2848. doi:10.2514/1.G009050.

[3] F. De Grossi, A. Carbone, D. Spiller, D. Ottaviani, R. Mengoni, C. Circi, Transcription and optimization of an interplanetary trajectory through quantum annealing, Astrodynamics 9 (2) (2025) 195–215. doi:10.1007/s42064-024-0216-6.

[4] G. L. Moretta, A. Casasola, Quantum-assisted initialization for low-thrust earth-moon trajectories, Zenodo presentation (2026). doi:10.5281/zenodo.18723794.
URL https://zenodo.org/records/18723794

[5] D. Morante, M. Sanjurjo Rivo, M. Soler, A survey on low-thrust trajectory optimization approaches, Aerospace 8 (3) (2021) 88. doi:10.3390/aerospace8030088.

[6] M. T. Ozimek, K. C. Howell, Low-thrust transfers in the earth-moon system, including applications to libration point orbits, Journal of Guid-

ance, Control, and Dynamics 33 (2) (2010) 533–549. doi:10.2514/1.43179.

[7] R. E. Pritchett, K. C. Howell, D. J. Grebow, Low-thrust transfer design based on collocation techniques: Applications in the restricted three-body problem, in: AAS/AIAA Astrodynamics Specialist Conference, Stevenson, WA, 2017, aAS 17-626.
URL https://engineering.purdue.edu/people/kathleen.howell.1/Publications/Conferences/2017_AAS_PriHowGre.pdf

[8] Y. Kayama, K. C. Howell, M. Bando, S. Hokamoto, Low-thrust trajectory design with successive convex optimization for libration point orbits, Journal of Guidance, Control, and Dynamics 45 (4) (2022) 623–637. doi:10.2514/1.G005916.

[9] C. Hofmann, F. Topputo, Rapid low-thrust trajectory optimization in deep space based on convex programming, Journal of Guidance, Control, and Dynamics 44 (7) (2021) 1379–1388. doi:10.2514/1.G005839.

[10] A. C. Morelli, A. Morselli, C. Giordano, F. Topputo, Convex trajectory optimization using thrust regularization, Journal of Guidance, Control, and Dynamics 47 (2) (2024) 339–346. doi:10.2514/1.G007646.

[11] Y. Sidhoum, K. Oguri, Indirect forward-backward shooting for low-thrust trajectory optimization in complex dynamics, Journal of Guidance, Control, and Dynamics 47 (10) (2024) 2164–2172. doi:10.2514/1.G007997.

[12] N. Kumagai, K. Oguri, Adaptive-mesh sequential convex programming for space trajectory optimization, Journal of Guidance, Control, and Dynamics 47 (10) (2024) 2213–2220. doi:10.2514/1.G008107.

[13] NASA Jet Propulsion Laboratory, Three-body periodic orbits api, accessed 14 June 2026 (2020).
URL https://ssd-api.jpl.nasa.gov/doc/periodic_orbits.html

[14] A. Rubinsztejn, C. G. Sandel, R. Sood, F. E. Laipert, Designing trajectories resilient to missed thrust events using expected thrust fraction, Aerospace Science and Technology 115 (2021) 106780. doi:10.1016/j.ast.2021.106780.

[15] C. Venigalla, J. A. Englander, D. J. Scheeres, Multi-objective low-thrust trajectory optimization with robustness to missed thrust events, Journal of Guidance, Control, and Dynamics 45 (7) (2022) 1255–1268. doi:10.2514/1.G006056.

[16] N. Kumagai, K. Oguri, Robust cislunar low-thrust trajectory optimization under uncertainties via sequential covariance steering, Journal of Guidance, Control, and Dynamics 48 (12) (2025) 2725–2743. doi:10.2514/1.G009092.

[17] J. Varghese, K. Oguri, Trajectory optimization under uncertainty with nonlinear programming and forward-backward shooting, Journal of Guidance, Control, and Dynamics 49 (1) (2026) 59–77. doi:10.2514/1.G009259.

[18] H. Sekine, K. Oguri, Y. Kawabata, R. Funase, Robust trajectory optimization against missed thrust events using sequential convex programming, in: 2026 IEEE Aerospace Conference, 2026, pp. 1–15. doi:10.1109/AERO66936.2026.11520058.

[19] N. Bernardini, N. Baresi, R. Armellin, State-dependent trust region for successive convex programming for autonomous spacecraft, Astrodynamics 8 (4) (2024) 553–575. doi:10.1007/s42064-024-0200-1.

[20] D-Wave Systems, Qubos and ising models, accessed 14 June 2026 (2026).
URL https://docs.dwavequantum.com/en/latest/quantum_research/qubo_ising.html

[21] F. Glover, G. Kochenberger, Y. Du, A tutorial on formulating and using qubo models (2018). arXiv:1811.11538.
URL https://arxiv.org/abs/1811.11538

[22] E. Farhi, J. Goldstone, S. Gutmann, A quantum approximate optimization algorithm (2014). arXiv:1411.4028.
URL https://arxiv.org/abs/1411.4028

[23] A. Shukla, P. Vedula, Trajectory optimization using quantum computing, Journal of Global Optimization 75 (1) (2019) 199–225. doi:10.1007/s10898-019-00754-5.

[24] NASA Space Place, How far away is the moon?, accessed 16 June 2026 (2026).
URL https://spaceplace.nasa.gov/moon-distance/

Table 1: Compact claim hierarchy for the manuscript. Pass/fail labels refer only to the configured normalized screening evidence, not to flight validation.

Claim	Primary artifacts	Status	Prohibited interpretations
Benchmark resource	Source states, configs, result CSV/JSON, generated tables/figures, cache metadata, and artifact manifest	Pass as an auditable normalized CR3BP resource	Operational targeting, flight certification, or mission-design completeness
Initializer comparison	30-seed raw results, success intervals, paired comparisons, threshold sensitivity, and QAOA/QUBO ablation tables	Negative for quantum/QAOA superiority; positive only for duty-cycle-aware refinability	Quantum advantage, hardware evidence, or sampled-method dominance over all-windows continuous refinement
Continuous-backend diagnostics	Continuation, direct collocation, Hermite–Simpson, replay/stress sidecars, and metadata packages	Mixed: several scoped CR3BP rows pass configured screens, while harder rows fail	Certified solver parity, fuel optimality, or universal missed-thrust robustness
Hard-catalog tail-coast	Tail-coast recovery CSV, accepted-control replay CSV/metadata, branch audit, and threshold audit	Scoped pass under normalized CR3BP for all 27 configured one/two masks	High-fidelity robustness, broader outage-family robustness, broad production-solver validation/parity, or quantum evidence
Force-model stress bridges	Cached Horizons contrast, independent-HS bicircular, cached-Horizons, and SPICE-derived replay CSV/metadata, independent-HS cached-Horizons point-mass retuning CSV/metadata, hard-catalog bicircular stress CSV/metadata, and generated tables	Contrast-only for hard-catalog Horizons; positive simple bicircular and cached-Horizons-derived stress replays for the converged independent-HS row; persisted controls fail direct point-mass replay but independent retuning restores feasibility; SPICE-derived replay preserves the retuned	Full high-fidelity SPICE trajectory propagation, ephemeris flight validation, accepted-control high-fidelity replay, broad production-solver validation/parity, or DOI/release proof

Table 2: Cross-backend evidence synthesis generated from recorded artifacts only. Error entries are nominal / selected-worst / all-mask-worst in the normalized terminal metric. Tighter pass statuses use all-mask diagnostics conservatively; selected-branch rows remain diagnostics unless all configured masks are explicitly selected.

Axis	Representative evidence	Role	Scope	Nom / Sel / All	Configured	Tighter or near-tight status	Practitioner lesson
30-seed threshold sensitivity	tight (0.050,0.090) pass counts	sampled binary initializers versus all-windows continuous	30 deterministic seeds; selected one-segment recovery masks	counts only	see source threshold table	sampled methods 0/30; all-windows 30/30	Tight screening exposes margin limits of sampled high-duty schedules; the positive result is the benchmark diagnostic, not sampler superiority.
continuation-extraction multiple shooting	all_single_p04_warm_from_p03	all one-segment continuous continuation from p=0.3	8/8 configured one-segment masks selected and evaluated	0.0531 / 0.0539 / 0.0539	True	0.065/0.10: True; 0.05/0.09: False	All-mask continuation has headroom at p=0.4 under 0.065/0.10 screening, but the nominal error is above 0.05.
continuation-extraction multiple shooting	two_segment_n8_p03_cold	cold all one/two-mask continuous continuation	15/15 configured one- and two-segment masks selected and evaluated	0.0612 / 0.0435 / 0.0435	True	0.065/0.10: True; 0.05/0.09: False	A broader one/two-mask phase-shift row passes 0.065/0.10 all-mask diagnostics, but not the 0.05 nominal headroom check.
direct collocation baseline	selected branch p=0.4	compact trapezoidal selected-branch collocation	1 selected outage branch; 12 one-segment all-mask diagnostics	0.0257 / 0.0591 / 0.0602	True	0.065/0.10: True; 0.05/0.09: True	Compact collocation can be tight on this selected branch, but unselected masks remain diagnostics and should be reported.
independent-midpoint Hermite-Simpson	dsc_phase_p03_amax02_warm_from_p03	selected-branch independent-midpoint collocation with tighter thrust	3 selected outage branches; 8 one-segment all-mask diagnostics	0.0197 / 0.0508 / 0.0532	True	0.065/0.10: True; 0.05/0.09: True	Independent midpoint controls meet tight all-mask diagnostics on this phase-shift row even at amax=0.2.
independent-midpoint Hermite-Simpson all-configured headroom	dsc_all_single_p04_amax02_warm_from_p03	all one-segment independent-midpoint collocation with tighter thrust	8/8 configured one-segment masks selected and evaluated	0.0531 / 0.0774 / 0.0774	True	0.065/0.10: True; 0.05/0.09: True	The independent-midpoint HS headroom package upgrades the earlier selected-branch row to all-configured one-segment evidence at p=0.4 and amax=0.2, while retaining the max-nfev optimizer caveat.
independent-HS branch-control replay	original and polish p=0.4, amax=0.2 branch sidecars	deterministic replay of persisted endpoint-plus-midpoint controls	16 branch replay rows; 8/8 one-segment masks for original and polish rows	/ 0.0774 / 0.0779	True	0.065/0.10: replay delta only; 0.05/0.09: replay delta only	Persisted independent-HS branch controls replay exactly under normalized CR3BP; this strengthens recovery-side reproducibility without adding high-fidelity or production-solver evidence.
independent-HS simple bicircular phase-sweep stress	dsc_all_single_p04_amax02_polish_from_p04	persisted endpoint-plus-midpoint controls replayed with simple circular solar tide	8/8 configured one-segment masks over Sun phases 0.0, 45.0, 90.0, 135.0, 180.0, 225.0, 270.0, 315.0	0.0221 / 0.0856 / 0.0856	True	0.065/0.10: True; 0.05/0.09: True	The converged independent-HS all-configured row has a positive deterministic beyond-CR3BP stress replay under the simple circular solar-tidal model; the result is useful benchmark-resource evidence, not flight validation.
independent-HS cached-Horizons-derived solar-tidal replay	dsc_all_single_p04_amax02_polish_from_p04	persisted endpoint-plus-midpoint controls replayed with cached JPL Horizons geometry	8/8 configured one-segment masks at representative 2026-Jan-01 epoch	0.0184 / 0.0742 / 0.0742	True	0.065/0.10: True; 0.05/0.09: True	This is stronger than the simple bicircular phase sweep because the Sun geometry comes from cached JPL Horizons vectors, but it remains a simplified stress replay, not SPICE, high-fidelity/flight validation or broad production-solver validation/parity.
independent-HS cached-Horizons Earth/Moon/Sun point-mass retuning	dsc_all_single_p04_amax02_polish_from_p04	independent endpoint-plus-midpoint retuning under cached-Horizons point-mass dynamics	8/8 configured one-segment masks at representative 2026-Jan-01 epoch	0.0214 / 0.0247 / 0.0247	True	0.065/0.10: True; 0.05/0.09: True	Persisted controls fail direct ephemeris point-mass replay, but independent retuning restores feasibility for nominal and all 8 branches at the representative epoch; this is not SPICE, full high-fidelity/flight validation or broad production-solver validation/parity.
independent-HS multi-epoch cached-Horizons Earth/Moon/Sun point-mass retuning	dsc_all_single_p04_amax02_polish_from_p04 over four 2026 epochs	independent endpoint-plus-midpoint retuning under cached-Horizons point-mass dynamics	4 representative 2026 epochs; 32 configured one-segment branch rows retuned	0.0214 / 0.0247 / 0.0247	True	0.065/0.10: True; 0.05/0.09: True	The cached-Horizons point-mass retuning result is no longer a single representative-epoch check: nominal direct replay fails in all four fixed 2026 epochs; direct branch replay passes 18/32 overall (July 8/8); retuning restores feasibility. This is still not SPICE, full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence.
independent-HS SPICE-derived ephemeris replay	dsc_all_single_p04_amax02_polish_from_p04 over four 2026 epochs	already-retuned endpoint-plus-midpoint controls replayed under SPICE-derived vectors	4 fixed 2026 epochs; 4 nominal and 32 branch controls replayed with no retuning	0.0214 / 0.0247 / 0.0247	True	0.065/0.10: True; 0.05/0.09: True	The multi-epoch point-mass retuned controls replay under SPICE-derived Moon/Sun vectors with sub-1e-9 normalized-error deltas relative to the cached-Horizons retuned replay. This strengthens ephemeris-source provenance but remains point-mass stress replay, not full high-fidelity/flight validation or broad production-solver validation/parity.
independent-HS scoped CasADi/IPOPT bridge check	dsc_all_single_p04_amax02_polish_from_p04	scoped mature NLP backend direct-shooting bridge check from accepted sidecars	1 nominal and 8 configured one-segment branch rows; branch outage segments fixed inactive; branch prefixes fixed to refined nominal masked controls; zero-variable branches=1	0.0091 / 0.0155 / 0.0155	True	0.065/0.10: True; 0.05/0.09: True	One normalized-CR3BP accepted independent-HS polish case exports to CasADi/IPOPT and locally refines under the same target/scales/branch masks while preserving source branch-recovery semantics. This is a scoped mature-backend portability/bridge check that addresses the production-solver-parity concern only in that narrow sense; it is not broad production-solver validation/parity, production mission design, high-fidelity/flight validation, global or fuel optimality, DOI evidence, or quantum evidence.
hard-catalog tail-coast recovery	tail_coast_all_one_two_segment_t5_portfolio	locked-nominal fixed-final-time tail-coast branch portfolio	27/27 configured one- and two-segment masks selected and evaluated	0.0230 / 0.0936 / 0.0936	True	0.05/0.10: True; 0.05/0.09: False	The hard-catalog pass is real but scoped to the tail-coast backend; it passes 0.05/0.10 and narrowly misses 0.05/0.09 because the selected/all worst error is [illegible].
independent-midpoint Hermite-Simpson	dsc_catalog_dro_tf4_amax1_selected1	bounded selected-outage independent-midpoint collocation negative diagnostic	1 selected outage branch; 15 one/two-mask all-mask diagnostics	4.2644 / 11.0843 / 11.0843	False	0.065/0.10: False; 0.05/0.09: False	The hard-catalog result is not a generic continuous-backend pass; this independent-HS selected-outage diagnostic fails badly.

Table 3: Fixed-final-time tail-coast locked-nominal hard-catalog recovery. The combined row selects/evaluates all 27 configured one- and two-segment masks in one fixed-final-time case row; separate all-single and all-two-segment rows are retained as provenance/scope rows. Branches with post-outage recovery variables are optimized. No-recovery-variable late-tail branches are direct threshold evaluations, so false branch optimizer all-success flags are accounting caveats rather than threshold failures.

Case	Policy	Outage lengths	Selected masks	Tail coast segments	Portfolio variants	Fallback eval branches	Accepted fallback branches	Eligible optimizer-converged branches	Direct no-recovery branches	Tail-coast nominal error	Selected fixed-time worst error	All-mask fixed-time diagnostic worst	Max $\|\|u\|\|$	Bound violation	nfev	Meets thresholds
tail_coast_nominal_only_t5	none	[1]	0	5	1	0	0	0/0	0	0.0230	0.0230	27.8351	1.0000	0.0000	287	True
tail_coast_all_single_t5_portfolio	all single	[1]	14	5	2	1	1	13/13	1	0.0230	0.0230	0.0230	1.0000	0.0000	2057	True
tail_coast_all_two_segment_t5_portfolio	all configured	[2]	13	5	2	3	3	12/12	1	0.0230	0.0936	0.0936	1.0000	0.0000	3467	True
tail_coast_all_one_two_segment_t5_portfolio	all configured	[1, 2]	27	5	2	4	4	25/25	2	0.0230	0.0936	0.0936	1.0000	0.0000	5929	True

Table 4: Thirty-seed $N = 12$ duty-cycle-prior phase-shift method comparison. Values summarize 210 method–seed rows. QUBO/QAOA totals include 96 shared QUBO-training evaluations and 24 post-QUBO true candidate evaluations.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Active target penalty	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0928	0.1188	0.1277	0.8333	0.0833	0.8333	0.8333	120.0000	0.0000	120.0000	0.3667
catalog_halo_phase_shift	cross_entropy	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	genetic	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0675	0.0926	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	true_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	24.0000	96.0000	120.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0948	0.1208	0.1299	0.8333	0.0833	0.8333	0.8333	24.0000	96.0000	120.0000	0.4333
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	111111111111	111111111111	111111111111	0.0386	0.0601	0.0690	1.0000	0.0833	1.0000	1.0000	19.0000	0.0000	19.0000	1.0000

Table 5: Thirty-seed main-method statistical diagnostics. Error deltas are method minus baseline for refined selected-worst error; negative values favor the method. Very small sign-test values are rendered in scientific notation rather than rounded to zero.

method	success	wilson_95_ci	delta_success_vs_all_windows_continuous	mean_error_delta_vs_all_windows_continuous	sign_p_vs_all_windows_continuous	delta_success_vs_surrogate_qubo_sa	mean_error_delta_vs_surrogate_qubo_sa	sign_p_vs_surrogate_qubo_sa
random	11/30	[0.22, 0.54]	-0.63	+0.0595 [+0.0533, +0.0659]	1.86e-09	-0.63	+0.0264 [+0.0201, +0.0328]	5.95e-05
cross_entropy	30/30	[0.89, 1.00]	+0.00	+0.0333 [+0.0327, +0.0337]	1.86e-09	+0.00	+0.0002 [-0.0006, +0.0009]	0.6291
genetic	30/30	[0.89, 1.00]	+0.00	+0.0325 [+0.0318, +0.0331]	1.86e-09	+0.00	-0.0006 [-0.0015, +0.0002]	0.3323
true_sa	30/30	[0.89, 1.00]	+0.00	+0.0334 [+0.0329, +0.0338]	1.86e-09	+0.00	+0.0003 [-0.0004, +0.0010]	0.8238
surrogate_qubo_sa	30/30	[0.89, 1.00]	+0.00	+0.0331 [+0.0326, +0.0335]	1.86e-09			
qaoa_statevector	13/30	[0.27, 0.61]	-0.57	+0.0484 [+0.0422, +0.0544]	1.86e-09	-0.57	+0.0153 [+0.0090, +0.0214]	0.1849
all_windows_continuous	30/30	[0.89, 1.00]				+0.00	-0.0331 [-0.0335, -0.0326]	1.86e-09

Table 6: Threshold-sensitivity counts recomputed from the recorded 30-seed raw CSV. A row passes when both `refined_nominal_error` and `refined_selected_worst_error` are below the listed thresholds. No trajectory optimization is rerun.

Threshold	(e_n, e_s)	Random	CEM	Genetic	True SA	QUBO SA	QAOA	All-windows
Configured	(0.090, 0.170)	11/30	30/30	30/30	30/30	30/30	13/30	30/30
Stricter	(0.075, 0.120)	4/30	30/30	30/30	30/30	30/30	13/30	30/30
Stringent	(0.065, 0.100)	3/30	2/30	4/30	1/30	2/30	8/30	30/30
Tight	(0.050, 0.090)	0/30	0/30	0/30	0/30	0/30	0/30	30/30

Table 7: Thirty-seed QAOA/QUBO statistical diagnostics. Error deltas are method minus baseline for refined selected-worst error; negative values favor the method.

method	success	wilson_95_ci	delta_success_vs_sa	mean_error_delta_vs_sa	sign_p_vs_sa	delta_success_vs_random_qaoa	mean_error_delta_vs_random_qaoa	sign_p_vs_random_qaoa
surrogate_qubo_sa	28/30	[0.79, 0.98]						
exact_qubo_top24	28/30	[0.79, 0.98]	+0.00	+0.0006 [-0.0026, +0.0039]	0.3323			
qaoa_random_p1	15/30	[0.33, 0.67]	-0.43	+0.0119 [+0.0053, +0.0184]	0.5716			
qaoa_optimized_p1	21/30	[0.52, 0.83]	-0.23	+0.0048 [-0.0012, +0.0109]	1.0000	+0.20	-0.0072 [-0.0141, -0.0001]	0.3616
qaoa_optimized_p2	29/30	[0.83, 0.99]	+0.03	-0.0006 [-0.0039, +0.0024]	0.4545	+0.47	-0.0125 [-0.0188, -0.0060]	0.2478

Table 8: Compact scientific lessons from the current evidence package.

Lesson	Evidence and boundary
Dense-control baseline	All-windows continuous refinement is the necessary dense-control baseline. It remains lower-error in paired selected-worst comparisons and still passes 30/30 seeds at the tight $(0.050, 0.090)$ threshold where every sampled method is 0/30.
Evidence semantics	Selected-branch evidence, all-mask diagnostics, and all-configured-mask evidence must be separated. A row is not all-configured evidence unless it explicitly optimizes/evaluates every configured mask in that scope.
CR3BP headroom	The all-configured independent-midpoint Hermite–Simpson package gives phase-shift CR3BP headroom for eight one-segment masks: the original p=0.4 row has nominal error 0.0111 and all-configured worst error 0.0774 but retains the `max_nfev`/`optimizer_success=False` caveat; the polished row converges with nominal error 0.0113 and worst error 0.0779. Persisted branch endpoint/midpoint controls replay exactly under normalized CR3BP. A scoped CasADi/IPOPT mature NLP backend bridge check for one normalized-CR3BP accepted case refines the accepted polish nominal/branch-worst errors to 0.00914/0.01553 with IPOPT success 9/9 while preserving branch-recovery semantics, but it is a local bridge check rather than production mission design or broad production-solver validation.
External validity	The converged independent-HS polish row passes a simple bicircular phase sweep with maximum nominal/branch errors 0.02214/0.08557 and a cached-Horizons-derived representative-epoch replay with errors 0.01836/0.07422 and branch pass count 8/8. Cached-Horizons Earth/Moon/Sun point-mass replay records persisted-control failure (0.38126 nominal); independent retuning restores 0.02144 nominal and 0.02473 branch-worst errors for all 8 branches at the January epoch, and the new four-epoch wrapper repeats the stress/retuning check across Jan/Apr/Jul/Oct 2026, where nominal direct replay fails in 4/4 epochs, direct branch pass count is 18/32 overall (July 8/8), and retuned branch pass count is 32/32. A SPICE-derived replay of those retuned controls keeps branch pass count 32/32 with maximum